

Session Objectives

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- Learning Objective 1-Identify the different types of aphasia.
- Learning Objective 2- Localize different stroke syndromes and the arteries that supply them.
- Learning Objective 3- Understand the difference between an arterial and a venous stroke.
- Learning Objective 4- Recognize the risk factors for venous infarcts.
- Learning Objective 5- Recognize the different types of intracranial hemorrhages.

Known Stroke Risk Factors

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Modifiable Risk Factors

- Tobacco use/Regular cannabis use
- Hypertension >130/80
- Dyslipidemia
- Diabetes
- Kidney disease
- Sleep apnea
- Heavy Etoh
- Diet
- Atrial Fibrillation

Non-modifiable Risk factors

- Male sex
- Age
- Genetic susceptibility
- Race



Source: Continuum Lifelong Learning in Neurology. Cerebrovascular Disease. Volume 26, Number 2. April 2020. Image: DW.COM

Questions to Consider

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Which side of the brain is language function generally localized to?

Why do neurologists always ask if a patient is right or left handed?

Transcortical Motor Aphasia

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- Stroke to left hemisphere, but spares the perisylvian cortex. Frontal lobe lesion.
- Nonfluent aphasia (hesitant speech)
- **Repetition intact**
- Good comprehension
- Anterior cerebral artery territory lesion.

Transcortical Sensory Aphasia

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- Analogue of Wernicke's receptive aphasia
- Poor comprehension
- **Intact repetition, may be echolalic**
- Sparing of perisylvian cortex
- Reading and writing is impaired

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A lateral view of the human brain with the superior temporal gyrus highlighted in red. The label 'superior temporal gyrus' is written in red text across the highlighted area.

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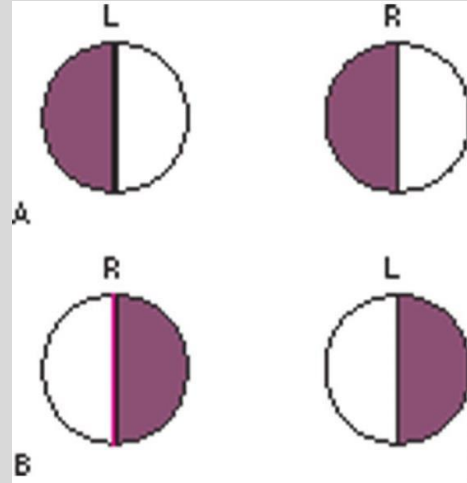
Posterior Cerebral Art Infarct

- PCAs are major terminal branches of the basilar artery. They supply branches to the midbrain, thalamus, occipital lobes and medial and inferior temporal lobes.
- Unilateral infarctions of occipital lobe may cause contralateral homonymous hemianopia

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Neglect

- Inability to process information on one side of the body.
- More severe after right sided cerebral lesions. May occur after **right posterior parietal stroke**.
- May treat a body part as foreign

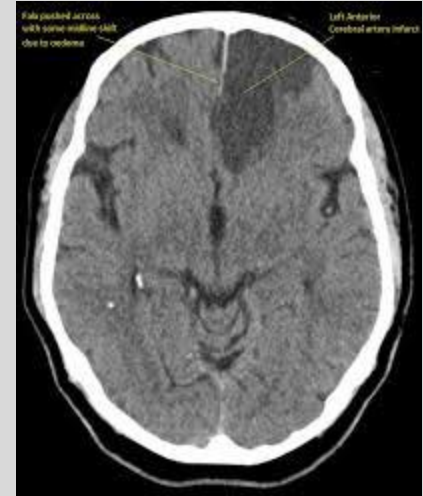


Anterior Cerebral Art. Infarct

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- Uncommon in isolation (<3%)
- ASA stroke can occur due to 2 main types of interruption to the brain's blood flow. 1) Blockage or clot 2) Hemorrhagic stroke due to aneurysm rupture. **Contralateral lower extremity > arm weakness.**
- Abulia, akinetic mutism impaired memory and emotional disturbances.



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Image; Neurovascularmedicine.com

Internal Carotid Artery Stroke

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- ACA and MCA are branches
- Atherosclerotic plaque/ critical stenosis at cervical bifurcation
- Contralateral hemiparesis and hemisensory loss (face>arm>leg) and motor>sensory
- Transient central retinal artery occlusion- better prognosis than hemispheric stroke.

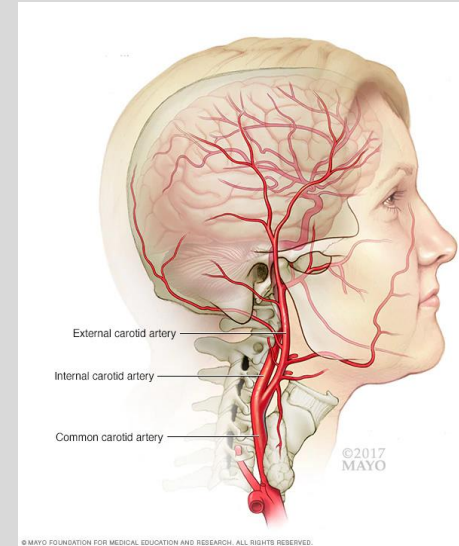


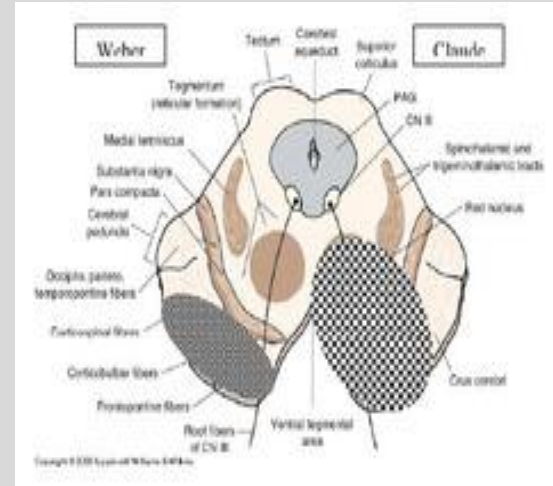
Image- mayo clinic
Umass Chan Medical School Stroke Stope

Brainstem Syndromes

- “Locked in”
- Wallenberg (lateral medullary)
- Medial medullary
- Weber Syndrome
- Foville Syndrome
- Benedikt Syndrome
- Claude Syndrome
- Nothnagel Syndrome

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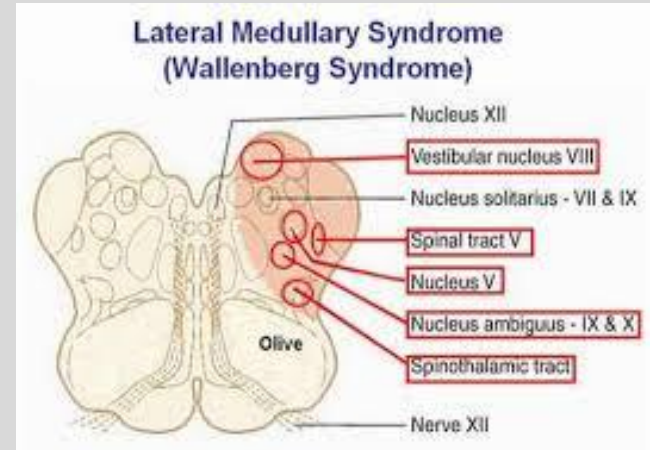


Lateral Medullary Syndrome

- Vertigo, nausea/vomiting, nystagmus
- Numbness- ipsilateral face and contralateral body
- Ipsilateral ataxia
- Ipsilateral paralysis of larynx, pharynx and loss of gag

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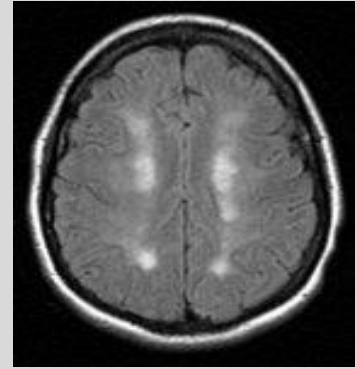


Man in a Barrel Syndrome

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- Anoxia to the cerebral border zones due to systemic hypoperfusion.
- Bilateral arm weakness while sparing the lower extremities.
- Damage to cortex that represents the arms



Lacunar infarcts

- Pure motor-post limb int. capsule
- Pure sensory- Ventral post lat nuc thalamus
- Mixed sensorimotor-thalamocapsular
- Dysarthria-clumsy hand syndrome-genu int cap, basis pontis
- Ataxia hemiparesis-basis pontis, internal capsule.

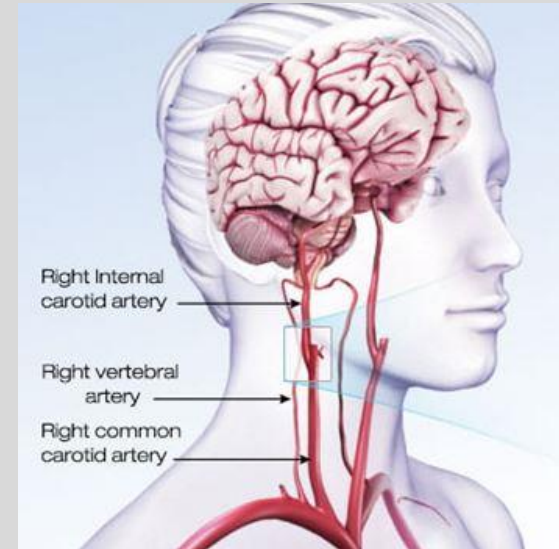
TIA

- A transient ischemic attack is a temporary blockage of blood flow to the brain.
- The clot usually dissolves on its own or gets dislodged. Symptoms usually last less than 5 minutes, but can last up to 24 hours.
- A TIA can help prevent a stroke, as it is a warning.

American Stroke Association

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Clinical challenge of TIA

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- Stroke mimics: Migraine, epilepsy, multiple sclerosis, peripheral nerve entrapment, brain hemorrhage (i.e. subdural hematoma).
- Motor and speech symptoms have a higher likelihood of brain ischemia as the cause of the symptoms.
- About 20% of patients presenting with TIAs will have a stroke within the next 90 days. Highest risk is within the next 24 hours.

Epidural hematoma

- Often traumatic with lens shaped appearance.
- Rupture of middle meningeal artery after skull fracture.
- A lucid interval before LOC (talk and die).

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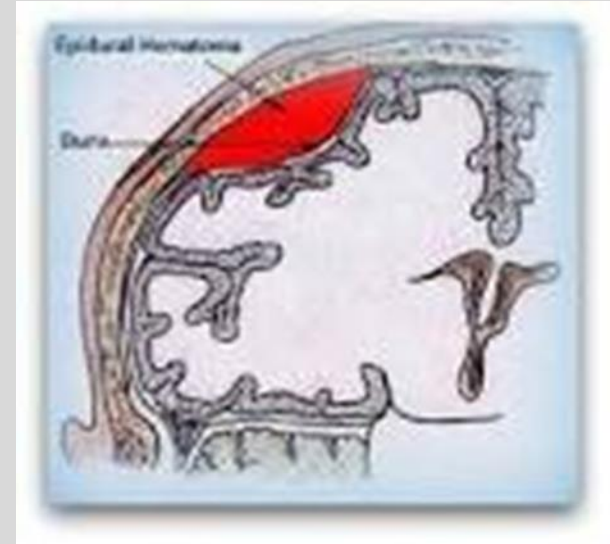


Image: Brain injury explanation. Continuum- Lifelong learning in Neurology. Cerebrovascular Disease. Volume 23, Number 1, February, 2017

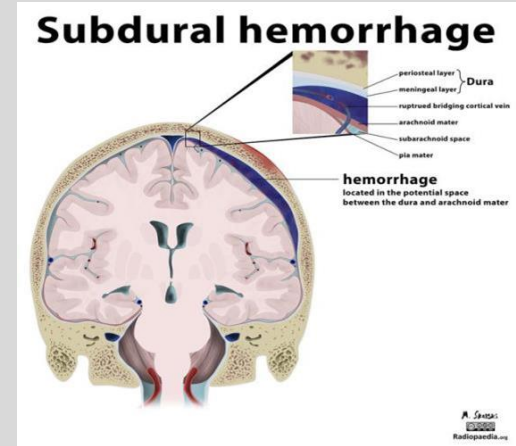
Subdural hematoma

- Usually due to trauma- MVAs, falls and assaults. A crescent shaped extra-axial lesion.
- Rupture of bridging veins
- Occurs between the dura and the arachnoid membranes.
- Trivial or minor head trauma in susceptible patients (cerebral atrophy).

Image- pixgalerie.com/mycleveandclinic.org
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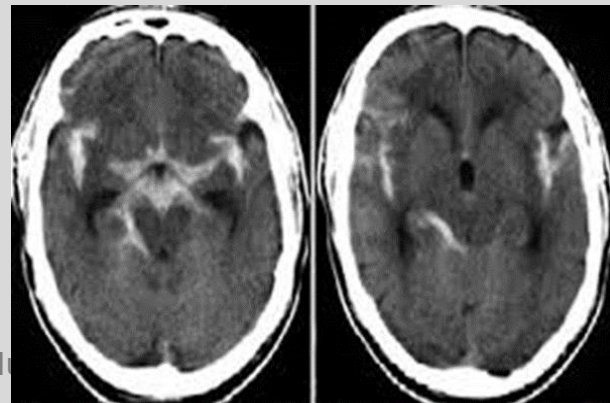


Subarachnoid Hemorrhage

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- Aneurysmal versus nonaneurysmal SAH
- Perimesencephalic is most common cause of **nonaneurysmal**.
 - Reversible cerebral vasoconstriction syndrome
 - Posterior reversible encephalopathy syndrome
 - Infective endocarditis, cocaine use
 - Cortical vein thrombosis
 - Coagulopathies



Case 5 -dissection

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- Jane is a 40-year-old woman who presented with a new severe headache on the right side of her head with associated neck pain. It began yesterday.
- “The worst headache I can remember.”
- History of thyroid cancer treated with radiation therapy.



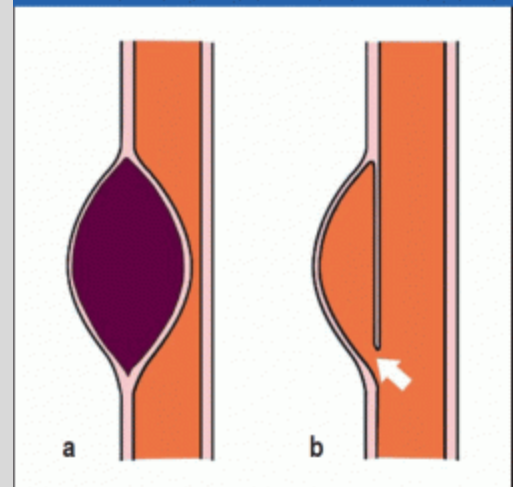
Arterial Dissection

- Blood collection and intramural hematoma formation in the arterial wall layers.
- May occur in the carotid artery, vertebral artery or both
- Most commonly involves the extracranial vessels
- Causes: Spontaneous or as a result of trauma.
- A common cause of stroke in young adults.

Causes of Cervical Dissection

- Connective tissue: Fibromuscular dysplasia, EDS IV, Polycystic Kidney
- Environmental factors: trauma (whiplash type)
- Use of oral contraceptives
- Migraine

FIGURE 1



Pathoanatomy of arterial dissection

a) Dissection of the internal carotid artery or vertebral artery with a hematoma in the vessel wall due to rupture of the vasa vasorum. The intramural hematoma can progress toward the lumen with secondary rupture of the intima.

b) Dissection of the aorta with rupture of the intima and formation of two lumina. Modified from (20).

Signs and Symptoms of Dissection

- New-onset head pain (ICA)
- Neck pain (vertebral artery)
- Ipsilateral Horner syndrome
- Cranial and cervical neuropathies
- Pulsatile tinnitus
- TIA or stroke may result

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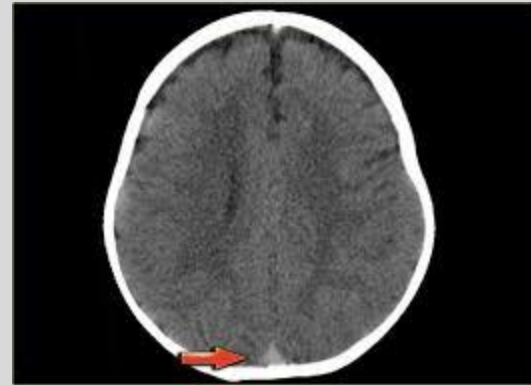


Venous Thrombosis

- Rates are higher than previously thought.
- Rates among women of reproductive age are consistently 3 times higher than their male counterparts.
- Associated conditions: genetic prothrombotic diseases, antiphospholipid syndrome, cancer, oral contraceptive use, infections (COVID-19)

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Clinical Syndromes of CVT

- Signs and symptoms are diverse. Presentation can be divided into four syndromes:
- 1) Headache (90%) or increased intracranial pressure
- 2) Focal neurologic presentations/ seizures
- 3) Subacute encephalopathy
- 4) Cavernous sinus syndrome with multiple cranial neuropathies.

Diagnosis of CVT

- Requires a high degree of suspicion. Cerebral angiogram is the gold standard, but noninvasive imaging is favored. CT and MR venography or MRI with contrast.
- **Hemorrhagic conversion seen in greater than 1/3 of patients (often parasagittal distribution).**

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